

VALIBOYINA MURALI SAI

✉ muralisaivaliboyina@gmail.com ☎ +91 9347110168 🔗 [linkedin.com/in/valiboyina-murali-sai-ba5689250](https://www.linkedin.com/in/valiboyina-murali-sai-ba5689250)

🐙 github.com/VALIBOYINA-MURALI-SAI

EXPERIENCE

Nokia

01/2025 – Present

Student Intern - Generative AI

- Built a hybrid RAG chatbot combining semantic and keyword search, improving retrieval precision by ~35% over single-method baseline.
- Implemented HNSW-based vector search, reducing query latency by ~60% compared to flat search.
- Designed agent-based workflows using LangChain to enable dynamic tool usage and multi-step reasoning.

Samsung PRISM

12/2023 – 05/2024

Research Intern

- Developed a web crawling pipeline to collect structured e-commerce data for LLM fine-tuning.
- Improved prompt generation relevance by ~22% using curated domain-specific datasets.
- Contributed to LLaMA-based model fine-tuning and evaluation workflow.

PROJECTS

HireSense AI

Python, Streamlit, spaCy, scikit-learn, SQLite3

- Built an AI-powered resume analyzer using TF-IDF and cosine similarity to match resumes with job descriptions.
- Used spaCy for entity extraction (skills, experience) to improve ATS scoring.
- Deployed using Streamlit Cloud with sub-2 second response time.

Gesture Genius

Python, OpenCV, MediaPipe, autopsy

- Developed a real-time hand gesture recognition system using 21-point hand landmarks.
- Achieved ~94% accuracy with <30ms latency using distance-based gesture classification.
- Enabled gesture-based mouse control and keyboard actions.

AI Meeting Summarizer

Python, Flask, Whisper, HuggingFace Transformers, NLTK

- Built an end-to-end pipeline: audio → transcription → summarization → structured output.
- Used Whisper for speech-to-text and BART for summarization.
- Reduced manual documentation time by ~70%.

EDUCATION

B.Tech - CSE (AI & ML)

SRM Institute of Science and Technology, Chennai

CGPA: 9.74 / 10

PUBLICATION

RAG-X: Density-Adaptive Path Sampling for Enhanced Knowledge Graph-Based Retrieval Augmented Generation

IEEE Conference Paper

- Proposed adaptive graph traversal to improve factual accuracy in RAG systems.
- Achieved 92.4% accuracy vs 65.8% baseline (+26.6%).
- Improved throughput by 4.9x using optimized retrieval + reranking.

SKILLS

AI / GenAI

LLMS, RAG, LangChain, PyTorch, TensorFlow

Data

NumPy, Pandas, Matplotlib, Plotly, scikit-learn, SQL

Tools

Git, Flask, Streamlit, Docker, Hugging Face Hub, Jenkins, SQL, LaTeX

CV/ NLP

OpenCV, MediaPipe, Hugging Face Transformers, spaCy, NLTK, Whisper, TF-IDF

Languages

Python, C++, C

ACHIEVEMENTS

Winner – SRM Hackathon 8.0

Built AI-based medical assistant with 98% accuracy fall detection

Ranked 4th in India - INTSO Mathematical Olympiad

Academic Excellence

- 98.2% (Class XII).
- 97.16% (Class X).

CERTIFICATIONS

AWS AI-ML & Google AI-ML – EduSkills

(Mar 2025)

Machine Learning in Urban Studies – ISRO IIRS

NPTEL: Intro to ML (IIT Madras), Data Mining Gold -Top 1% (IIT Kharagpur)

Languages Known

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|-----------|---------|
| • English | • Hindi |
| • Telugu | |